Diffusion and Score-Based Generative Models

Score Matching, Reverse SDEs, Probability-Flow ODEs, and Sampling Error

Thanh Nguyen Quoc

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Contents

Abstract

Diffusion and score-based generative models transform data into noise through a forward stochastic process and then learn to reverse this process. Their mathematical foundation involves score functions, denoising score matching, stochastic differential equations, Fokker-Planck equations, reverse-time dynamics, and numerical sampling. This report presents the main objects and highlights research questions concerning score approximation, statistical error, optimization error, and sampling error.

1. Forward Noising Process

A diffusion model defines a forward process that gradually corrupts data into noise. In continuous time, a common formulation is an SDE

$$dX_t = f(X_t, t)dt + g(t)dW_t, \qquad t \in [0, T],$$

where W_t is Brownian motion, f is a drift, and g is a diffusion coefficient. The marginal density of X_t is denoted by p_t.

The goal is to choose the forward process so that p_T is close to a simple reference distribution, often a standard Gaussian.

2. Score Function

The score function of a density p_t is

$$s_t(x) = \nabla_x \log p_t(x).$$

Score-based generative modeling trains a neural network $s_\theta(x, t)$ to approximate $s_t(x)$. The learned score is then used to reverse the noising process.

3. Reverse-Time SDE

Under appropriate regularity assumptions, the reverse-time dynamics can be written as

$$dX_t = \left[f(X_t, t) - g(t)^2 \nabla_x \log p_t(X_t)\right] dt + g(t) d\bar{W}_t,$$

where time is run backward and $\bar{W}_t$ is a reverse-time Brownian motion. In practice, the true score is replaced by the learned score s_θ.

Thus the sampler depends critically on the accuracy of score estimation.

4. Fokker-Planck Equation

The density p_t of the forward SDE satisfies a Fokker-Planck equation:

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g(t)^2 \Delta p_t.$$

This PDE viewpoint is important because it connects stochastic processes, density evolution, and diffusion-based generative modeling.

5. Probability-Flow ODE

For certain diffusion models, there exists a deterministic probability-flow ODE with the same time marginals:

$$\frac{dX_t}{dt} = f(X_t, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(X_t).$$

This ODE provides a deterministic sampler and connects diffusion models to continuous normalizing flows.

6. Denoising Score Matching

Directly estimating $\nabla \log p_t$ is difficult. Denoising score matching uses perturbed data. If

$$X_t = \alpha_t X_0 + \sigma_t Z, \qquad Z \sim \mathcal{N}(0, I),$$

then the conditional score is often available:

$$\nabla_{x_t} \log p(x_t | x_0) = -\frac{x_t - \alpha_t x_0}{\sigma_t^2}.$$

The model is trained by minimizing

$$\mathcal{L}(\theta) = \mathbb{E}_{t,x_0,z} \left[\lambda(t) \left\| s_\theta(x_t, t) + \frac{x_t - \alpha_t x_0}{\sigma_t^2} \right\|^2 \right].$$

7. Error Sources

Diffusion theory must account for several errors:

1. score approximation error: the neural network class is limited;
2. statistical error: training uses finite data;
3. optimization error: empirical loss is not minimized exactly;
4. time-discretization error: the reverse SDE/ODE is solved numerically;
5. initialization error: terminal distribution may not be exactly Gaussian;
6. model misspecification: assumptions on smoothness or support may fail.

A full guarantee should control how these errors propagate to the generated distribution.

8. Research-Level Error Decomposition

A typical objective is to prove a bound of the form

$$d(\rho_T^\theta, \mu) \leq C_1 \varepsilon_{\text{score}} + C_2 \varepsilon_{\text{stat}} + C_3 \varepsilon_{\text{opt}} + C_4 \varepsilon_{\text{disc}} + C_5 \varepsilon_{\text{init}},$$

where d may be total variation, Wasserstein distance, or another distributional metric. The constants and rates depend on regularity, dimension, noise schedule, and numerical solver.

9. Diffusion Alignment as Distributional Optimization

In alignment problems, one may start from a base diffusion model with generated distribution ρ_0, and seek an improved distribution ρ minimizing

$$\mathbb{E}_\rho[\ell(x)] + \lambda \mathrm{KL}(\rho \| \rho_0).$$

This formulation separates distributional preference optimization from neural-network parameterization. The mathematical questions include existence, KKT conditions, dual formulations, and sampling from the optimized distribution.

10. Research Directions

Important directions include:

1. statistical guarantees for score matching without exact empirical minimization;
2. Wasserstein convergence under non-smooth data distributions;
3. score estimation on low-dimensional manifolds;
4. primal-dual constrained diffusion alignment;
5. sampling error bounds for probability-flow ODE solvers;
6. stability of the generated distribution under perturbations of the score.

11. Exercises and Projects

1. Derive the conditional Gaussian score formula.
2. Simulate an Ornstein-Uhlenbeck process.
3. Implement Euler-Maruyama for a simple SDE.
4. Train a toy denoising score network on a Gaussian mixture.
5. Compare stochastic reverse-SDE sampling and deterministic probability-flow ODE sampling in a toy setting.
6. Write a research note on optimization, statistical, and sampling errors in diffusion models.

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